

Data

CGWB quarterly
well observations

Training-well
ordinary kriging

RF / HGBR
downscaling

Downscaled surface
(companion paper v1)

RL environment

Behaviour policies
random · greedy · ϵ -greedy

Synthetic simulator
8×8 blocks · quarterly

30,000-transition
offline dataset

Training

Behavioural
cloning (BC)

Conservative
Q-learning (CQL)

Evaluation

FQE
seed bootstrap

Direct rollout
episode bootstrap

Banded OOD
4 Mahalanobis bands

\$5.2
FQE ≠ rollout

